

Міністерство освіти і науки України
Національний університет «Запорізька політехніка»
Кафедра програмних засобів

ЗВІТ

Дисципліна «Системи автоматизованого проєктування»

Робота №2

Тема «Програмне формування креслеників»

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Прийняли

Викладач

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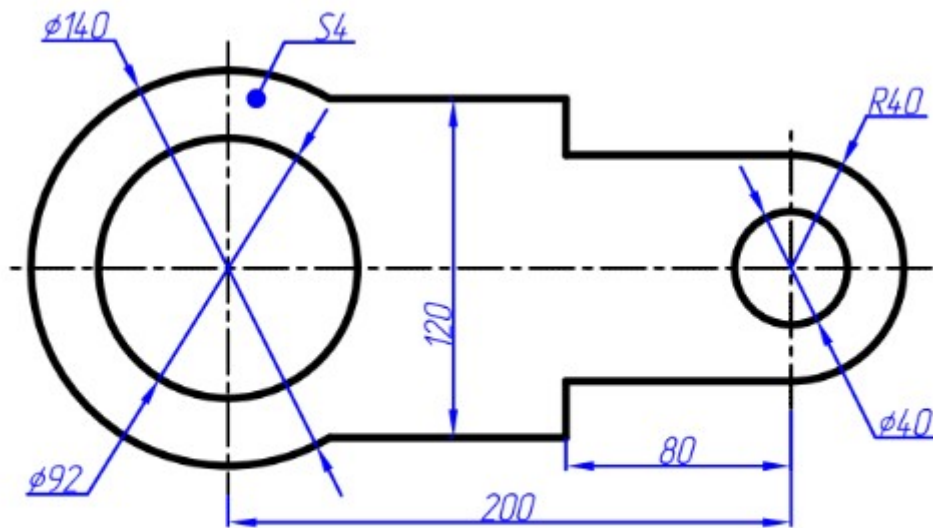
2025

МЕТА

На прикладі запропонованого елемента навчитися:

1. Працювати в середовищі AutoCAD у командному режимі;
2. Створювати пакетні файли команд (скрипти);
3. Програмно формувати кресленики за допомогою засобів мови AutoLISP.

ВИКОНАННЯ



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Рисунок 1.0 — Кресленик індивідуального завдання

Код програми SCR

```
grid_OFF
OSNAP_OFF
SNAP_OFF
LWDISPLAY_1
LINETYPE_S_CENTER__LWEIGHT_0.3
COLOR_40
LINE_0,75_0,-75_
LINE_200,45_200,-45_
LINE_-75,0_245,0_
```

```

LINETYPE_S_BYLAYER__LWEIGHT_0.5
COLOR_white
ZOOM_All
CIRCLE_0,0_D_92
CIRCLE_0,0_D_140
CIRCLE_200,0_D_40
CIRCLE_200,0_D_80
PLINE_200,40_120,40_120,60_0,60_
MIRROR_110,60__200,0_400,0_N
TRIM_-48.3743,49.6739__20,60_20,-60_
TRIM_70,60_70,-60__63.1369,29.7627_161.2380,10.5113_
LINETYPE_S_BYLAYER__LWEIGHT_0.3
COLOR_140
dimtxt_7
dimasz_7
dimdiameter_34.1107,30.3610_-70,-50
dimdiameter_-48.3743,49.6739_-40,80
dimdiameter_182.6092,9.5745_250,-40
dimradius_239.0539,-10.1209_240,40
dimhorizontal_120,-60_200,-45_200,-70
dimhorizontal_200,-70_0,-75_200,-90
dimvertical_120,60_120,-60_270,60

```

Код програми AutoLISP

```

(defun MainLayer ()
  (defun MainLayer ()
    (command "LINETYPE" "S" "BYLAYER" "" "LWEIGHT" 0.5 "" "COLOR"
"white")
  )
)

(defun AxisLayer ()
  (command "LINETYPE" "S" "CENTER" "" "LWEIGHT" 0.3 "" "COLOR" 40)
)

(defun SizeLayer ()
  (command "LINETYPE" "S" "BYLAYER" "" "LWEIGHT" 0.3 "" "COLOR" 140)
)

(defun c:gasket ()
  ; axes
  (setq top_left_vertical_axis_point (list 0 75))
  (setq bottom_left_vertical_axis_point (list 0 -75))
  (setq top_right_vertical_axis_point (list 200 45))
  (setq bottom_right_vertical_axis_point (list 200 -45))
  (setq left_horizontal_axis_point (list -75 0))
  (setq right_horizontal_axis_point (list 245 0))
  ; dimensions
  (setq select_left_inner_circle (list 34.1107 30.3610))
  (setq select_left_outer_circle (list -48.3743 49.6739))
  (setq left_inner_circle_dimension_point (list -70 -50))
  (setq left_outer_circle_dimension_point (list -40 80))
  (setq select_right_inner_circle (list 182.6092 9.5745))
  (setq select_right_outer_circle (list 239.0539 -10.1209))
)

```

```

(setq right_inner_circle_dimension_point (list 250 -40))
(setq right_outer_circle_dimension_point (list 240 40))
(setq select_left_eighty_dimension_point (list 120 -60))
(setq eighty_horizontal_dimension_point (list 200 -70))
(setq two_hundred_horizontal_dimension_point (list 200 -90))
(setq select_top_vertical_dimension (list 120 60))
(setq select_bottom_vertical_dimension (list 120 -60))
(setq vertical_dimension_point (list 270 60))

(command "grid" "OFF")
(command "OSNAP" "OFF")
(command "SNAP" "OFF")
(command "LWDISPLAY" 1)
(AxisLayer)
(command "LINE" top_left_vertical_axis_point
bottom_left_vertical_axis_point "")
(command "LINE" top_right_vertical_axis_point
bottom_right_vertical_axis_point "")
(command "LINE" left_horizontal_axis_point
right_horizontal_axis_point "")
(MainLayer)
(command "ZOOM" "All")
(command "CIRCLE" (list 0 0) "D" 92)
(command "CIRCLE" (list 0 0) "D" 140)
(command "CIRCLE" (list 200 0) "D" 40)
(command "CIRCLE" (list 200 0) "D" 80)
(command "PLINE" (list 200 40) (list 120 40) (list 120 60) (list 0
60) "")
(command "MIRROR" (list 110 60) "" (list 200 0) (list 400 0) "N")
(command "trim" select_left_outer_circle "" (list 20 60) (list 20
-60) "")
(command "trim" (list 70 60) (list 70 -60) "" (list 63.1369 29.7627)
(list 161.2380 10.5113) "")
(SizeLayer)
(command "dimtxt" 7)
(command "dimasz" 7)
(command "dimdiameter" select_left_inner_circle
left_inner_circle_dimension_point)
(command "dimdiameter" select_left_outer_circle
left_outer_circle_dimension_point)
(command "dimdiameter" select_right_inner_circle
right_inner_circle_dimension_point)
(command "dimradius" select_right_outer_circle
right_outer_circle_dimension_point)
(command "dimhorizontal" select_left_eighty_dimension_point
bottom_right_vertical_axis_point eighty_horizontal_dimension_point)
(command "dimhorizontal" eighty_horizontal_dimension_point
bottom_left_vertical_axis_point two_hundred_horizontal_dimension_point)
(command "dimvertical" select_top_vertical_dimension
select_bottom_vertical_dimension vertical_dimension_point)
)

```

РЕЗУЛЬТАТ

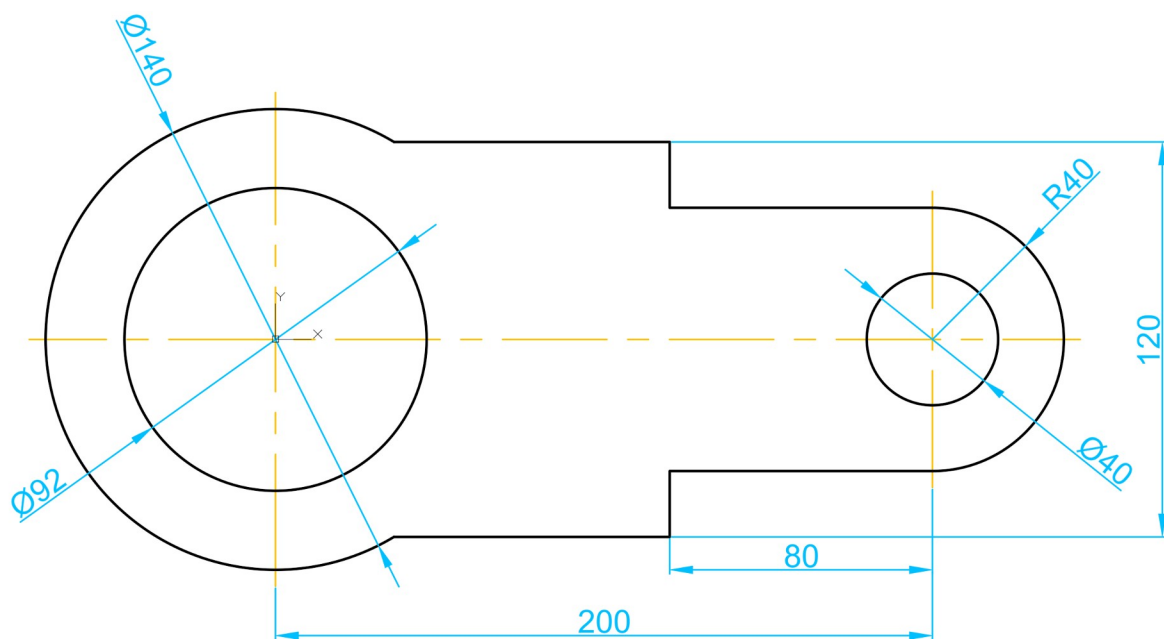


Рисунок 1.1 — Отриманий кресленик